Originality report

COURSE NAME
CSA4704

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1 of 7 passages

Student passage CITED

The sigmoid maps any real input into the interval (**0, 1**). Its derivative is never greater than 0.25 and approaches…

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The sigmoid function is a smooth S-shaped function that **maps any real input into** a value between **0** and **1**: σ(x) = 1 / (1 + e−x) Its key properties include: 0 < σ(x) < 1 σ(0) = 1/2 σ(x) → 0 as

The sigmoid function is a smooth S-shaped function that maps any … https://www.facebook.com/61578305660118/posts/the-sigmoid-function-is-a-smooth-s-shaped-function-that-maps-any-real-input-into/1221950954849435 22/

2 of 7 passages

Student passage FLAGGED

Let θ_t denote all model **parameters** at update **t**, $g_t = \nabla\theta L_t$ the current mini-batch gradient, **η the learning rate, and ε a small constant added for numerical stability**

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Update **Parameters**: θt+1=θt−(ηgt/E[g2]**t**+ε) **η** is **the learning rate and ε** is **a small constant added for numerical stability** (typically ε=10^−8). ...

Deep Learning — Mini Batch Gradient | by Krushna Thakkar - Medium https://medium.com/operations-research-bit/deep-learning-mini-batch-gradient-3bd1b7244e3f

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3 of 7 passages

Student passage FLAGGED

Momentum maintains **a velocity that accumulates a fraction of the previous update** direction. It can accelerate progress when gradients point in…

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RMSProp with **momentum** adds a completely separate mechanism: **a velocity** buffer **that accumulates a fraction of the previous** velocity plus the current (already adaptively rescaled) **update**, smoothing the…

What Is RMSProp? Root Mean Square Propagation Explained https://www.articsledge.com/post/root-mean-square-propagation-rmsprop

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4 of 7 passages

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RMSProp maintains **an exponentially** weighted **average of squared gradients** and **divides** each **gradient by the square root of** that scale. Parameters with consistently large gradients receive relatively…

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RMSProp keeps **an exponentially** decaying moving **average of** each parameter's **squared gradients**, then **divides** that parameter's current **gradient by the square root of** the average (plus a small stability…

What Is RMSProp? Root Mean Square Propagation Explained https://www.articsledge.com/post/root-mean-square-propagation-rmsprop

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5 of 7 passages

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…flattening, followed by a dense classifier. For ten classes, **the final layer produces** ten **logits; softmax converts them** to class **probabilities**.

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In the context of **the** model: Hidden layers learn features → **final layer produces logits** → **Softmax converts them** into **probabilities**. Without Softmax, the model outputs numbers.

Bhautik Prajapati's Post - LinkedIn https://www.linkedin.com/posts/bhautik-prajapati_what-softmax-actually-is-and-why-it-appears-activity-7450049652634198016-9QjC

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6 of 7 passages

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The receptive field is the region of the original **image that can influence** one feature-map **value**. Track r_l, the receptive-field width, and j_l, the distance…

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The receptive field of a unit in a convolutional neural network (CNN) **is the region of the** input **image that can influence** that unit's **value**. A unit in a network's first 3 × 3 convolution layer sees a…

Receptive Field: What a CNN Neuron Sees - Ultralytics https://www.ultralytics.com/glossary/receptive-field

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7 of 7 passages

Student passage FLAGGED

the current gradient; momentum smooths direction using a velocity; RMSProp divides by **a moving average of** squared gradients; **Adam** combines moving **first and second moments with bias correction**

Top web match

This is not **the** same thing as Adam's first-moment tracking, even though both involve **a moving average of** gradient-derived quantities -- RMSProp's momentum, when enabled, is applied after the adaptive…

What Is RMSProp? Root Mean Square Propagation Explained https://www.articsledge.com/post/root-mean-square-propagation-rmsprop

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